

GEOSPATIAL TECHNOLOGIES (GIS)

The Geospatial Technologies (GIS) Track is a 9-month professional diploma designed to prepare trainees for careers in GIS, geospatial software development, and spatial data analysis. The program combines computer science fundamentals with advanced geospatial technologies, covering GIS, remote sensing, spatial databases, Python, web application development, cloud technologies, and GeoAI. Through extensive hands-on training and real-world projects, trainees gain the technical and professional skills needed to design, develop, deploy, and manage modern geospatial solutions that meet industry demands.

JOB PROFILES

GIS Developer · GIS Analyst · GIS Engineer · GeoAI Engineer · GIS Solution Engineer · GIS Sales Engineer · GIS Db Administrator · GIS Specialist · GIS Instructor



TOTAL HOURS

1,227hrs

Theory + hands-on labs



DELIVERY

Blended

Online & on-site



UNIVERSITY GRADE

Good

Minimum grade required



Where graduates work

Esri North Africa HERE Technologies Qualit Standard (Qsit)
Etisalat Misr WE Vodafone Egypt Khatib & Alami
Dar Al-Handasah Schneider Electric Elsewedy Hassan Allam
Edge-Pro DSC Ministry of Planning Ministry of Agriculture DST



Who may apply

Applicants with a minimum grade of Good or higher are welcome from the following disciplines:

- Computer Engineering
- Architecture Engineers
- Urban Planner Engineers
- Civil Engineers
- Geologists
- Petroleum Engineer



Selection process

01 IQ & English Exam

- Online exam covering problem-solving aptitude and English proficiency.

02 Technical Exam

- Computer-based assessment covering the program's core technical fundamentals.

03 Technical Interview

- Interview with the technical committee.

04 Interpersonal Interview

- Candidates who pass the technical interview are invited to this final stage.

Program content – 1,227 hours

CATEGORY	HOURS	SHARE	KEY COURSES
Fundamental Courses	288 hrs	23%	Computer Networks Fundamentals, Operating Systems Fundamentals, Database Fundamentals, Introduction to Programming, Object-Oriented Programming, Data Structures & Algorithms, Client-Side Technologies, HTML5 & CSS3, Introduction to Generative AI and Prompt Engineering, Vibe Coding & Working with Gen-AI APIs, Advanced Gen-AI Programming (RAG & Agentic Software)
Core Courses	582 hrs	47%	GIS Fundamentals using ArcGIS, Geo-Spatial Analysis, Remote Sensing & Digital Image Processing, Remote Sensing Applications (ArcGIS Pro & ERDAS Imagine), Geo-Apps Development (MERN & .NET), GeoAI, Geospatial Servers & Cloud, Python in GIS, Open Source GIS
Soft Skills	57 hrs	5%	Communication Essentials for Professionals, High Impact Presentations, Job Seeking Skills, Professional Demeanor
Graduation Project	300 hrs	24%	

The program combines instructor-led lectures, hands-on GIS labs, AI-powered learning, and self-study modules through MaharaTech to build practical skills using ArcGIS Pro, QGIS, Python, Web GIS, Cloud, and GeoAI technologies

Student Deliverable

The program prepares graduates for globally recognized professional certifications while equipping them with the practical skills required for successful careers in the geospatial industry

Before you apply

Explore these resources before your interview. Applicants are expected to have reviewed this material.

- Career Road Map for GIS "on Maharatech"
<https://maharatech.gov.eg/enrol/index.php?id=39>
- GIS Analyst and GIS Developer Job Profiles on Maharatech
<https://maharatech.gov.eg/course/view.php?id=1393>
- Learn more about GIS, Location Intelligence and ESRI technologies
<https://www.esri.com/en-us/location-intelligence/overview>
- Introduction to GIS Mapping "on Coursera"
<https://www.coursera.org/learn/introduction-gis-mapping/home/welcome>
- ITI values and vision
<http://www.iti.gov.eg/Site/AboutUs>

APPLY AT

ITI.GOV.EG

Applications open each intake period. Check the website for current dates and deadlines.